

Youdahe Asfaw

(612) 469-7001 | youdaheasfaw@gmail.com | [linkedin.com/in/youdahe](https://www.linkedin.com/in/youdahe) | github.com/youdahe | youdahe.com

EDUCATION

Minnesota State University Mankato

Bachelor of Science in Computer Science

Expected Graduation - June 2028

3.5/4.0

EXPERIENCE

Software Engineer Intern

Aug 2026 – Nov 2026

NMDP

Minneapolis, MN

- Decomposing a legacy C#/.NET monolith's federal data allocation/reporting domain into domain-bounded Java microservices over gRPC and Kafka, collapsing tangled orchestration into bounded service contracts.
- Prototyping payload compaction and async pipelines to hold sub-second processing latency under high concurrent load, deployed on AWS with containerized services, baselining the current system to isolate where time and cost accrue.

Software Engineer Intern

May 2026 – Aug 2026

Liberty Mutual Insurance

Portsmouth, NH

- Built an internal self-service app (Node.js, Bash/Linux) replacing manual sign-off from 20 engineers with cron-based probes tracking node availability and connection state; cut AIX decommissioning from 2–3 weeks to 1 day and retired 15+ nodes.
- Migrated on-prem workloads to Kubernetes using StatefulSets, PVC-backed storage, and CNI pod networking, working across an 8-engineer platform team under senior code review.

Software Engineer Intern

Jan 2026 – May 2026

Medica

Minnetonka, MN

- Shipped a Java/Spring document ingestion pipeline to production over 50K+ internal documents with a 9-person team, covering chunking, embedding generation, and Qdrant vector indexing with metadata filtering.
- Cut onboarding task time from 1 week to 2 days with hybrid sparse-dense retrieval and cross-encoder reranking, scoring 95% precision and recall on a 100+ conversation LLM-as-judge eval set.

Machine Learning Research Assistant

Feb 2025 – Jun 2025

Gustavus Adolphus College

Saint Peter, MN

- Cut form processing time 75% (60 to 15 minutes) with a voice-conditioned completion pipeline (LayoutLMv3, Azure Form Recognizer, ElevenLabs) resolving field semantics from joint textual, spatial, and visual layout structure.
- Sole researcher under two faculty advisors; generalized extraction across languages and input types at 85% field accuracy, presented at Macksey (Johns Hopkins), PRISM (UChicago), and two national symposia.

Software Engineer Intern

Aug 2023 – Aug 2024

Medtronic

Mounds View, MN

- Built the ingestion service layer for a centralized data platform consolidating 500K+ entries from fragmented legacy sources (Excel inventories, YAML configs, device telemetry, event logs, MP4 diagnostics), exposing internal APIs that normalized heterogeneous payloads into a canonical schema behind a repository layer.
- Decoupled ingestion from downstream processing with async batched writes, idempotency keys, and retry logic preventing duplicate writes on transient failure; added structured logging, health checks, and integration tests across API and persistence boundaries.

PROJECTS

Docere | *TypeScript, React, Node.js, SQL, Docker, Qdrant*

- Built an AI tutor for Minnesota districts with no CS coursework — embeds every student question and lesson completion into a vector store, so it answers with full context of the student's history and alerts teachers when someone starts falling behind.
- Grew to **500+ students across partner districts and raised \$25,000 to fund the platform.**

youdaheDB | *Rust, Raft, CRDTs*

- Built a database engine from scratch in Rust with no external libraries — a log-structured storage layer that writes to disk safely, survives crashes without losing data, merges its own files in the background, and handles concurrent readers and writers, tested by killing the process mid-write and benchmarking under load.
- Invited to write on youdaheDB for Algorythm (7,000+ subscribers):** “Building a Distributed Database From Scratch With Rust.”

Distributed Event Streaming Platform | *Go, Linux*

- Built a Kafka-style commit log with append-only partitioned segments, per-partition monotonic offsets, deterministic key-based producer routing, and leader/follower replication acknowledged at a configurable durability level.
- Implemented tiered storage rolling immutable hot segments to object storage with retained lookup metadata, plus consumer-group rebalancing, bounded-buffer backpressure, and idempotent offset commits; instrumented consumer lag, throughput, and p99 latency in Prometheus/Grafana.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go, Rust, Python, Java, SQL

Frameworks: FastAPI, React, LangChain/LangGraph, Node.js, Spring

Developer Tools: Docker, Kubernetes, AWS, PostgreSQL, Redis, Qdrant, Prometheus/Grafana, Git, Bash/Linux

ADDITIONAL

Activities & Organizations: National Society of Black Engineers (NSBE), ColorStack, CodePath, Management Leadership for Tomorrow (MLT), Genesys Works